Ejercicios

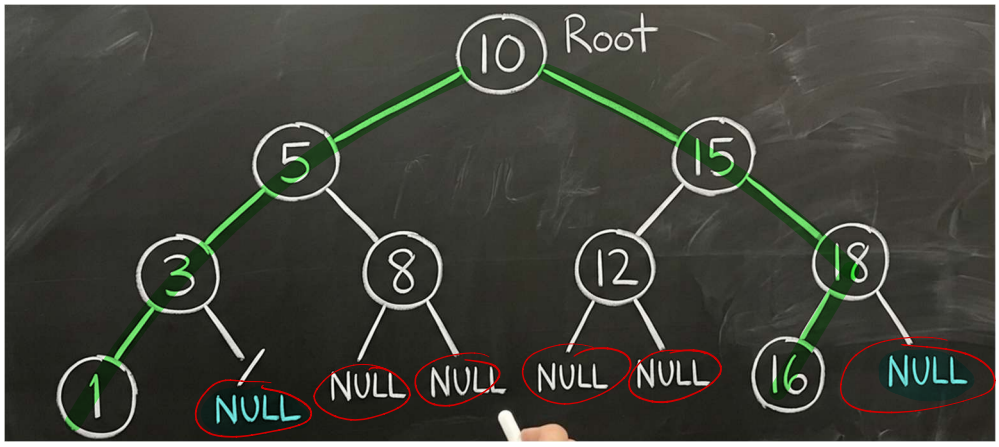
Para cada árbol determine:

1. la altura del árbol,
2. si es completo,
3. si está lleno y
4. si es balanceado
5. Altura y profundidad de:
    a. Nodo 1,
    b. Nodo 3,
    c. Nodo 5,
    d. Nodo 10

Haga recorrido:

1. Pre-Order
2. In-Order
3. Post-Order

1.



1) altura -> 3

2) completo ? -> no

3) lleno ? -> no

4) balanceado -> si

5) Altura y profundidad

Nodo 1 -> 0, 3

Nodo 3 -> 1, 2

Nodo 5 -> 2, 1

Nodo 10 -> 3, 0

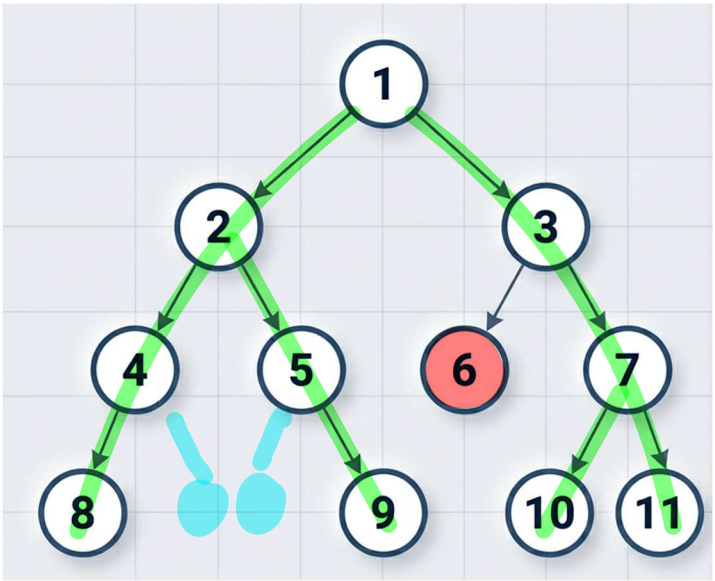
6) recorrido

Pre-Order -> 10, 5, 3, 1, 8, 15, 12, 18, 16

In-Order -> 1, 3, 5, 8, 10, 12, 15, 16, 18

Post-Order -> 1, 3, 8, 5, 12, 16, 18, 15, 10

2.



1) altura -> 3

2) completo ? -> no

3) lleno ? -> no

4) balanceado -> si

5) Altura y profundidad

Nodo 1 -> 3, 0

Nodo 3 -> 2, 1

Nodo 5 -> 1, 2

Nodo 10 -> 0, 3

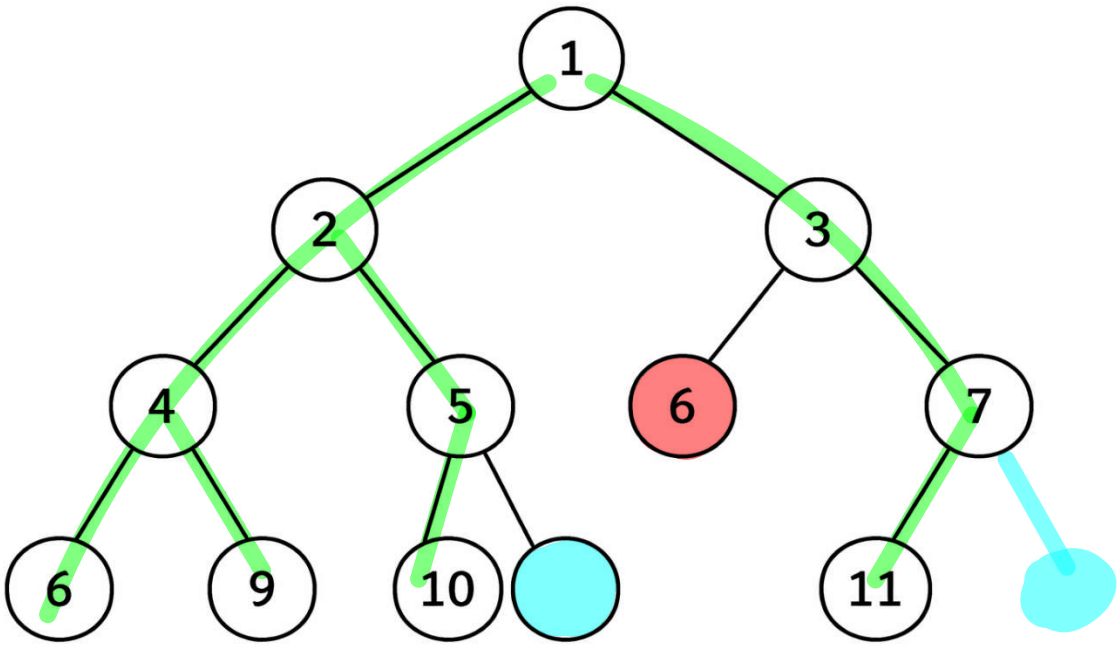
6) recorrido

Pre-Order -> 1, 2, 4, 8, 5, 9, 3, 6, 7, 10, 11

In-Order -> 8, 4, 2, 5, 9, 1, 6, 3, 10, 7, 11

Post-Order -> 8, 4, 9, 5, 2, 6, 10, 11, 7, 3, 1

3.



1) altura -> 3

2) completo ? -> no

3) lleno ? -> no

4) balanceado -> si

5) Altura y profundidad

Nodo 1 -> 3, 0

Nodo 3 -> 2, 1

Nodo 5 -> 1, 2

Nodo 10 -> 0, 3

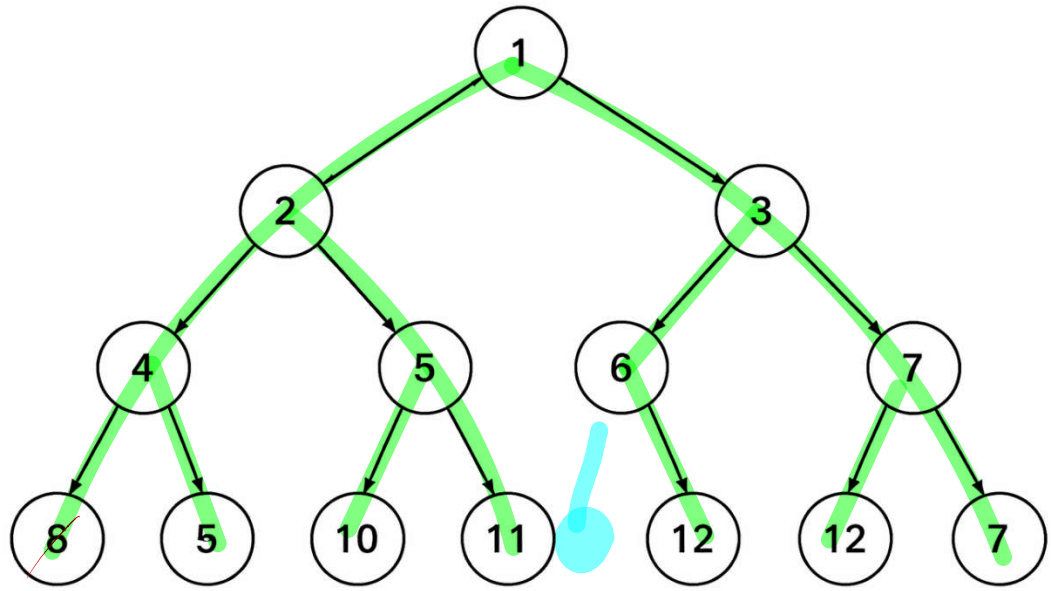
6) recorrido

Pre-Order -> 1, 2, 4, 6, 9, 5, 10, 3, 6, 7, 11

In-Order -> 6, 4, 9, 2, 10, 5, 1, 6, 3, 11, 7

Post-Order -> 6, 9, 4, 10, 5, 2, 6, 11, 7, 3, 1

4.



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1) altura -> 3

2) completo ? -> si

3) lleno ? -> no

4) balanceado -> si

5) Altura y profundidad

Nodo 1 -> 3, 0

Nodo 3 -> 2, 1

Nodo 5 -> 0, 3 / 1, 2

Nodo 10 -> 0, 3

6) recorrido

Pre-Order -> 1, 2, 4, 8 5, 5 ,10 ,11, 3, 6, 12, 7, 12, 7

In-Order -> 8, 4, 5, 2, 10, 5, 11, 1, 6, 12, 3, 12, 7, 7

Post-Order -> 8, 5, 4, 10, 11, 5, 2, 12, 6, 12, 7, 7, 3, 1